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MILITARY STANDARD

DATA STORAGE TRANSPARENCIES FOR FACTORED TRANSPARENCY RADAR SIMULATORS

/FSC 6940/

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DEPARTMENT OF THE NAVY

U. S. NAVAL TRAINING DEVICE CENTER

PORT WASHINGTON, NEW YORK

Data Storage Transparencies for Factored Transparency
Radar Simulators
MIL-STD-786(ONR/NTDC)

1. This standard has been approved by the Office of Naval Research (U.S. Naval Training Device Center), and is published to establish the naval standard practices for data storage transparencies used in the factored transparency system of radar simulation.

2. Recommended corrections, additions, or deletions should be addressed to the Commanding Officer and Director, U.S. Naval Training Device Center (Code 3301), Port Washington, New York.

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FOREWORD

The factored transparency system of radar simulation utilizes a pair of photographic images on a glass plate or plates to store topographic and radar reflection data. The data is scanned by a flying-spot scanner cathode ray tube, and the density of the images is read by two photomultiplier tubes. The two planar dimensions of the two images are the x and y dimensions of the topographic and reflectance data respectively. The densities of the reflectance images are used to store the intrinsic strength of radar target reflectance. The images are identical in their x and y values but separated in one dimension by the optical spacing of the dual readout system.

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CONTENTS

		Page
1.	SCOPE	1
1.1	Scope	1
1.2	Classification	1
1.2.1	Type I	1
1.2.2	Type II	1
2.	REFERENCE DOCUMENTS	1
3.	DEFINITIONS	2
3.1	Terms used	2
3.1.1	Transmittance	2
3.1.2	Reflectance	2
3.1.3	Density	2
3.1.4	Gamma	2
3.1.5	Sayce chart	2
3.1.6	Edge gradient	2
4.	GENERAL REQUIREMENTS	3
4.1	Transparency data	3
4.1.1	Characteristic curve	3
4.1.2	Radar reflectance data	3
4.1.3	Elevation data	3
4.2	Area data	3
4.2.1	Special cartography	4
4.3	Transparency numbering	4
4.3.1	Area designation	4
4.3.2	Successive copies	4
4.3.3	Modification designation	4
4.3.4	Engraving	4
4.3.5	Number assignment	5
5.	DETAIL REQUIREMENTS	5
5.1	Transparency dimensions	5
5.2	Simulated area	5
5.3	Map projection	5
5.3.1	Register marks	5
5.3.2	Elevation-transmittance relationship	5
5.4	Processing control	5
5.4.1	Densities	5
5.4.2	Gray scale control	5
5.4.3	Emulsion-gray tone reproduction	6
5.4.3.1	Emulsion resolution	6
5.4.3.2	Emulsion granularity	6
5.4.4	Elevation-transmittance slope accuracy	6
5.4.5	Elevation-transmission values, accuracy	6
5.5	Register marks, position	6
5.5.1	Transparency registration	6
5.5.2	Type II transparency assembly	6

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Check the source to verify that this is the current version before use.

CONTENTS

		Page
5.6	Type I transparency positioning	7
5.6.1	Transparency positioning adjustment	7
5.7	Transparency flatness	7
5.8	Transparency discontinuities	7
5.9	Radar detail	7
5.9.1	Category I	7
5.9.1.1	Cultural targets	7
5.9.1.1.1	Populated places	8
5.9.1.1.2	Bridges	9
5.9.1.1.3	Railroad marshaling yards	9
5.9.1.1.4	Causeways	9
5.9.1.1.5	Embankments	9
5.9.1.1.6	Areas bordering highways	10
5.9.1.1.7	Breakwaters	10
5.9.1.1.8	Dams	10
5.9.1.1.9	Airfield runways	10
5.9.1.1.10	Mines, tailings, and dumps	10
5.9.1.1.11	Quarries	10
5.9.1.1.12	Walls	10
5.9.1.1.13	Pipelines	10
5.9.1.2	Hydrographic features	11
5.9.1.2.1	Perennial lakes	11
5.9.1.2.2	Nonperennial lakes	11
5.9.1.2.3	Reservoirs	11
5.9.1.2.4	Waterways	11
5.9.1.2.5	Shorelines	11
5.9.2	Category II	11
5.9.2.1	Cultural targets	12
5.9.2.1.1	General	12
5.9.2.1.2	Bridges	12
5.9.2.2	Hydrographic features	12
5.9.2.2.1	Lakes	12
5.9.2.2.2	Waterways	12
5.9.3	Low altitude effects	12
5.10	Elevation-transmittance relationships	13
5.10.1	Elevation data	13
5.10.2	Elevation data, type I transparencies	13
5.10.2.1	Elevation-transmittance slopes	13
5.10.2.2	Lower coordinates	13
5.10.2.3	Upper coordinates for letter slope	14
5.10.2.4	Upper coordinates for numbered slope	14
5.10.2.5	Elevation-transmittance curves	14
5.10.3	Elevation data, type II transparencies	14

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Check the source to verify that this is the current version before use.

CONTENTS

		Page
5.10.3.1	Elevation-transmittance relationship.....	14
5.10.3.2	Density value of increments.....	14
5.10.3.3	Elevation value assignment.....	14
5.11	Radar return values.....	14
5.12	Transparency case.....	15

FIGURES

Figure 1.	Factored Transparency Data Plate	16
2.	Type II Factored Transparency Data Plates-Ass'y	17
3.	Type I Transparency Plate Positioning	18

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1. SCOPE

1.1 Scope.- This standard covers the configuration, density patterns, and dimensions for the photographic transparencies used as a map data storage device in the factored transparency system of radar simulation.

1.2 Classification.- Two configurations of factored transparencies are defined herein as follows:

1.2.1 Type I.- This transparency is a photographic image on a glass plate 16 by 20 by 1/4 inches, contains the elevation and reflectance images, and covers a geographic area of 510 by 1,315 nautical miles. It is detailed on figure 1.

1.2.2 Type II.- This transparency is a photographic image on a pair of glass plates, each 18 by 18 by 1/4 inches. One plate contains the elevation data image while the other contains the reflectance data image. The transparencies cover a geographic area of 1,250 by 1,250 nautical miles and are assembled in a fixed relationship by the use of a frame as detailed in figure 2.

2. REFERENCE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

DRAWINGS

U.S. Naval Training Device Center

640426	Type II Transparency Frame
640427	Type II Transparency Assy
640435	Spacer Flat
640436	Retainer, Film Plate
640437	Screw, Registration, Film Plate

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.- The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION

PH2.19-1959 American Standard for Diffuse Transmission Density

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, N.Y., 10016)

3. DEFINITIONS

3.1 Terms used.- The following are definitions of terms used in this standard:

3.1.1 Transmittance.- Transmittance is the percent of light transmitted by a transparency, relative to the amount of light impinging on it and measured according to Standard ASA PH2.19-1959.

3.1.2 Reflectance.- Reflectance refers to the percent of energy reflected by an object relative to the amount of energy impinging on it. The term is used herein in connection with both the intensity of return energy of radar from targets and with the intensity of return of light from a pattern of gray tones.

3.1.3 Density.- Reflection or transmission density is the logarithm, to the base 10 ($\log_{10}$), of the reciprocal of the reflectance or transmittance.

3.1.4 Gamma.- The slope of the "straight" portion of a plot of photo material of density vs. $\log_{10}$ exposure producing it.

3.1.5 Sayce chart.- This chart is manufactured only by W.&L.E. Gurley of Troy, New York. The 1958 Gurley master shall only be used to produce the 300-line charts, occupying approximately 2-1/8 by 5-3/4 inches, for the purposes of this standard.

3.1.6 Edge gradient.- Edge gradient is defined as the function $\frac{\Sigma G^2 X}{N}$ where $G = \Delta D/\Delta X$ (D is density, X is the measurement interval, and N is the number of measurements) greater than 5.

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4. GENERAL REQUIREMENTS

4.1 Transparency data.- The following data, describing a given factored transparency, shall be contained in a folder appropriately identified, and shall be shipped with each transparency.

4.1.1 Characteristic curve.- The characteristic curve of the process used to produce both the reflectance and elevation transparencies shall be plotted by the Herter and Driffield (D log E) method, to insure that additional copies of the transparency may be reproduced from the same masters at any time.

4.1.2 Radar reflectance data.- The measured transmittance for each level of radar return shall be furnished with the reflectance transparency.

4.1.3 Elevation data.- The measured transmittance for each elevation increment shall be furnished with the elevation transparency. This data shall be tabulated and shall also be plotted on linear graph paper where the X ordinate is the transmittance and the Y ordinate is the elevation given in feet.

4.2 Area data.- The following data shall also be contained in the folder specified in 4.1:

(a) The scale of the transparency

(b) The date on which the master compilation of the transparency was completed

(c) The latitude and longitude of each register point

(d) The indicated local bearing from the southerly or westerly register point to the other register point

(e) The distance in scale nautical miles between the two register points

(f) The transparency shall be based upon a Lambert Conic Conformal Projection which has standard parallels such that 60 percent of the area of the transparency falls between the parallels

(g) The scale size in feet of the minimum information bit used throughout the transparency

(h) The name, longitude and latitude, and the area coverage of all special target prediction data used in the transparency; and the degree of reliability and completeness of the prediction data

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(i) The date on which all basic data such as air target charts (ATC) or other special prediction data for target radar predictions were obtained

4.2.1 Special cartography.- Where cartography is based upon more than one Lambert Conic Conformal Projection, or for some other reason the cartography exceeds the bounds of the Lambert Conic Conformal Projection System, the following data shall be contained in the folder specified in 4.1.

(a) Special notes regarding the accuracy or completeness of the source maps

(b) The latitude of the boundary between each of the conic projections used to make the transparency, when more than one projection is used

(c) Limitations on the use of the transparency

4.3 Transparency numbering.-

4.3.1 Area designation.- A 3-digit number shall be used on each transparency to designate the area covered as follows:

(a) Numbers beginning with 100, European areas

(b) Numbers beginning with 200, Asian areas

(c) Numbers beginning with 300, North American areas

(d) Numbers beginning with 400, South American areas

(e) Numbers beginning with 500, African areas.

4.3.2 Successive copies.- Successive copies shall be identified by adding a lower case letter, starting with "a", to designate each copy of a particular master map. (Example: 302a, 302b, etc.)

4.3.3 Modification designation.- Modification designation shall be added by using Roman numerals starting with "I" to designate each modification of the master map. (Example: Number 302bII will refer to the second copy of the second modification to area 302)

4.3.4 Engraving.- The transparency number shall be marked on the transparency assembly in accordance with figure 1, for type I transparencies or figure 2 for type II transparencies. The designation of the master registration plate used to align type II transparencies in the mounting frame shall also be marked on the transparency assembly.

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4.3.5 Number assignment.- The transparency manufacturer shall apply to the U.S. Naval Training Device Center, Port Washington, N.Y., 11050, Attention Code 51, for the sequential number for any given transparency.

5. DETAIL REQUIREMENTS

5.1 Transparency dimensions.- The transparencies shall be as follows:

(a) Type I transparencies shall conform to the dimensions and configuration of figure 1

(b) Type II transparencies shall conform to the dimensions and configuration of figure 2

5.2 Simulated area.- The dimensions of figures 1 and 2 provide for a maximum geographic area, at a scale of 1:5,000,000, as follows:

(a) Type I--510 by 1,315 nautical miles

(b) Type II--250 by 1,250 nautical miles

5.3 Map projection.- The transparency and the master map shall be made using appropriate Lambert Conic Conformal Projection.

5.3.1 Register marks.- Register marks shall appear on all transparencies. Register marks for type I transparencies are detailed in figure 1, and for type II transparencies are detailed in figure 2.

5.3.2 Elevation-transmittance relationship.- The designation of the governing elevation-transmittance relationship shall be engraved on the transparency in accordance with figures 1 or 2 as appropriate.

5.4 Processing control.- Identical step wedges shall be placed on all transparencies. Type I transparency step wedges shall be in accordance with figure 1. Type II transparency step wedges shall be in accordance with figure 2. All step wedges shall contain a density identical to each elevation increment on the elevation image and each radar return code of the reflectance image.

5.4.1 Densities.- All densities shall be measured with a densitometer conforming to Standard ASA PH2.19-1959.

5.4.2 Gray scale control.- A copy of the laboratory characteristic curve and the corresponding reflectance-elevation control relationship, upon which the reflectance of each master map is based, shall be permanently attached to each master map. When a processing laboratory is required to copy an existing master map, the laboratory shall adjust its characteristic curve to duplicate the original characteristic curve as produced by the original laboratory.

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5.4.3 Emulsion-gray tone reproduction.- The emulsion used to image the master map shall be capable of the reproduction of gray values between 2 and 97 percent reflectance to a gamma or contrast function of 1, when measured from the characteristic curve data as specified by 4.1.1.

5.4.3.1 Emulsion resolution.- The emulsion shall be capable of reproducing 36 lines per millimeter at full amplitude when a sample is contact-printed from a Sayce resolution chart (see 3.1.5) using a diffuse light source so the density of the lines is 1.0 when the resultant print is measured with a microdensitometer reading linear density units and using a 5 by 150 μ aperture.

5.4.3.2 Emulsion granularity.- The standard deviation of its granularity (ΔD) shall be less than 0.0335 at 1.58 density and 0.0192 at 0.32 density when measured on a microdensitometer using a 48 μ aperture.

5.4.4 Elevation-transmittance slope accuracy.- The transmittance slope, plotted as specified in 4.1.3, shall fall within parallel lines separated by a value equal to 0.07 density. The slope shall not vary from the selected elevation-transmittance curve, in displacement or nonparallelism by more than a value equal to plus or minus 0.035 density.

5.4.5 Elevation-transmission values, accuracy.- The transmission value for any given assigned elevation shall have an accuracy of plus or minus one-half of the local elevation increment.

5.5 Register marks, position.- The position of the register marks shall be established within an accuracy of $\pm$ 500 scale feet. The register marks shall be noted with commensurate accuracy in seconds of latitude and longitude.

5.5.1 Transparency registration.- Ninety-five percent of all corresponding points on the reflectance and elevation transparencies shall register to within 0.00075 inch (300 feet at the transparency scale factor of 1:5,000,000). The remaining 5 percent shall register to within 0.0015 inch.

5.5.2 Type II transparency assembly.- The type II images shall be assembled in a frame as detailed in figure 2 and with the registration accuracy required by 5.5.1. The frame, necessary adhesive, and potting compound shall be in accordance with drawings 640426, 640427, 640435, 640436 and 640437.

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5.6 Type I transparency positioning.- The transparency shall be positioned in the simulator according to the arrangement and dimensions of figure 3. The bearing surfaces shown in figure 3 shall be used in all cameras used to produce transparencies and shall also exist in the transparency frame or holder of all surfaces.

5.6.1 Transparency positioning adjustment.- The positioning arrangement shall be such that a given transparency, exposed in such a jig, may be returned to a similar jig so that any point on the transparency is returned to the same position, relative to the jig, within $\pm$ 0.0025 inch. The simulators shall contain a means to correct misplacement of the image within a maximum of $\pm$ 0.003 inch. The placement correction system of the simulators plus the jigs shall be such that any point on the transparency can be repositioned to an accuracy of $\pm$ 0.0005 inch.

5.7 Transparency flatness.- The glass used for transparencies shall have a flatness of 0.0005 inch per linear inch or an overall flatness of 0.00035 inch per linear inch multiplier by either dimension of the glass.

5.8 Transparency discontinuities.- If the final transparency is fabricated utilizing separate sections or map panels, the change in transmittance across the boundaries of these panels shall be within the transmittance tolerance of that particular gray scale in accordance with 5.10 and 5.11.

5.9 Radar detail.-

5.9.1 Category I.- The requirements specified in RM-380, U.S. Air Target Charts, series 200/100, including the new RM-280 Interim High/Low Altitude Amendment, and RM-381 and RM-382, Area Radar Prediction Analyses (ARPA) shall be used as a guide in determining the radar data to be encoded on the factored transparencies.

5.9.1.1 Cultural targets.- The description of radar return features contained in this section is based upon the publications cited in 5.9.1 and is included herein in connection with the delineation of the densities in 5.11 by which these features shall be represented under various circumstances.

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5.9.1.1.1 Populated places.- Populated places include urban areas and are defined as a concentration of structures designed and built for human occupancy or human occupational activities. These structures vary in size from a few buildings to metropolitan areas. All populated places shall be portrayed to reflect the physical shape of the radar significant features of the area as far as possible in accordance with the following criteria:

(a) The radar significant features of populated places, in accordance with ATC series 100/200. The minimum dimensions of a target return area shall be 250 by 250 feet

(b) Any break in continuity of cultural build-up which is 400 feet or larger with a sigma of ± 50 feet measured on a root-mean-square (rms) basis

(c) Landmark buildings, such as large sheds, roofed storage areas, tanks, tank farms, missile launch sights, and the like, which are radar significant according to surface material and predominant height analysis, and have a minimum size of 250 by 250 feet:

(1) All predicted returns coded for return value shall be shown in their proper return value as delineated on the ATC series 200 or other source data specified by the pertinent training device specification

(2) All visually significant isolated buildings indicated on ATC by a solid black symbol and annotated shall be shown as determined by an analysis of the descriptive annotation

(3) All visually significant isolated buildings indicated on ATC by a solid black symbol and identified, shall be shown as one level above the surround

(4) All visually significant isolated buildings indicated on ATC by a solid black symbol and not identified shall not be shown

(5) Where a solid black symbol on ATC is surrounded by a first intensity return it shall be rendered as a first intensity return and merged with the surround

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(6) Other isolated targets which are not sufficiently large as specified in (a) above, but which are radar significant due to height and surface material, shall be portrayed at the minimum size of 250 by 250 feet

5.9.1.1.2 Bridges.- Steel reinforced concrete, or masonary bridges which are 500 feet or more in length shall be portrayed. Such bridges shall be shown with a maximum width of 250 feet. Steel bridges shall be encoded as return level number (No.) 1. Reinforced concrete or masonry bridges shall be encoded as return level No. 2. Lacking information as to the construction material, all open construction bridges shall be encoded as return level No. 1 and all other bridges shall be encoded as return level No. 3.

5.9.1.1.3 Railroad marshaling yards.- Railroad marshaling yards which have a minimum length of 1,000 feet and are at least 5 tracks wide shall be portrayed. No attempt shall be made to show tracks. Sidings or railyards which lie adjacent to each other shall be portrayed collectively.

5.9.1.1.4 Causeways.- Railroad or highway causeways which are 15 feet or more above surrounding water or marshlands and 1,000 feet or more long shall be portrayed. When the source data does not assign a specific reflectivity value, these causeways shall be encoded as land.

5.9.1.1.5 Embankments.- Embankments bordering highways or railroads when the embankments have a minimum height of 15 feet above the surrounding terrain and a minmum length of 1,000 feet shall be portrayed. When the source data does not assign a specific relfectivity value, these embankments shall be encoded as return level No. 4.

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5.9.1.1.6 Areas bordering highways.- The built-up cultural areas bordering highways approaching major towns or cities shall be portrayed.

5.9.1.1.7 Breakwaters.- Breakwaters which have a minimum length of 1,000 feet and a minimum height of 10 feet above the water, when the source data does not assign a specific reflectivity value, shall be encoded as land.

5.9.1.1.8 Dams.- Dams which have a minimum length of 200 feet and a minimum height on the downstream side of 10 feet shall be portrayed. Nonannotated dams shall be encoded as return level No. 3.

5.9.1.1.9 Airfield runways.- Airfield runways which are 5,000 feet or more in length and 200 feet or more in width shall be portrayed. The associated buildings, such as hangars, shall be shown when these meet the requirements for populated places. When the source data does not assign a specific reflectivity value, airfield runways shall be encoded with the reflectivity value of water.

5.9.1.1.10 Mines, tailings, and dumps.- Metal strip mines, tailings, and slag dumps which have a minimum height of 15 feet above the predominant height of the surrounding terrain structures and a minimum length of 1,000 feet shall be portrayed. When the source data does not assign a specific reflectivity value, mines, tailings and dumps shall be encoded as No. 1 special effects.

5.9.1.1.11 Quarries.- Quarries which have minimum dimensions of 500 by 500 feet shall be portrayed. When the source data does not assign a specific reflectivity value, quarries shall be encoded as No. 1 special effects.

5.9.1.1.12 Walls.- Prominent walls in open country which have a minimum height of 15 feet above the predominant surrounding terrain and a minimum length of 1,000 feet shall be portrayed. When the source data does not assign a specific reflectivity value, these walls shall be encoded as No. 1 special effects.

5.9.1.1.13 Pipelines.- Metal pipelines which are located above ground and are radar significant shall be portrayed. When the source data does not assign a specific reflectivity value, these pipelines shall be encoded as No. 1 special effects and be portrayed as a solid line 250 feet wide.

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5.9.1.2 Hydrographic features.-

5.9.1.2.1 Perennial lakes.- Perennial lakes which have a minimum dimension of 2,500 feet by 1,000 feet shall be portrayed. Adjacent swamp or marsh areas shall be encoded as marsh.

5.9.1.2.2 Nonperennial lakes.- Nonperennial lakes with a minimum dimension of 5,000 feet by 1,000 feet, when due to size, shape, contrast with the surroundings and the like, are considered to be an important radar identification point; in which case the maximum or most probable shoreline shall be shown.

5.9.1.2.3 Reservoirs.- Reservoirs with a minimum dimension of 2,500 feet by 1,000 feet shall be portrayed. The shoreline shall represent the water surface at the normal stage of the reservoirs as controlled by the spillway of the dam.

5.9.1.2.4 Waterways.- Rivers, navigable canals, major aqueducts, and flumes comprising major drainage patterns shall be encoded on the map. Such waterways shall be defined as waterways shown as double lines on the ATC 200/100 series or Army Map Service (AMS) series 250 charts. All such waterways having a width of 750 feet or greater, including adjacent swamp or marsh areas, shall be encoded as water return areas. For the purpose of generating appropriate waterway encoding data for transparency production from ATC and MAS charts, the above statements shall be interpreted as follows:

5.9.1.2.4.1 All waterways portrayed as double-line drainage on the ATC 200/100 or AMS series 250 charts shall be shown. When such waterways enter adjacent swamp or marsh areas they will be shown at a minimum width of 750 feet. Adjacent swamp or marsh shall be encoded as marsh.

5.9.1.2.5 Shorelines.- Shorelines in tidal waters shall be delineated as the outline of natural coastal features at mean high water. In inland waters, shorelines shall be mapped to correspond to the normal state of water.

5.9.2 Category II.- For the areas not covered by the ATC series 200/100, the targets listed below shall be included as radar significant. Any data, provided by the procuring agency prior to data freeze date, which normally would be expected to be shown on future revised AMS series 250 charts shall be, where possible, incorporated in the factored transparencies.

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5.9.2.1 Cultural targets.-

5.9.2.1.1 General.- The cultural targets shall be as shown on the AMS series 250 charts. In addition, target breakup shall be stored on the factored transparencies to the extent that data, such as major roadways, power transmission lines, major building complexes, and other radar significant features, are available.

5.9.2.1.2 Bridges.- Any bridge shown on the AMS series 250 charts which meets the size and structure requirements specified in 5.9.1.1.2 shall be provided. Lacking other information, bridges shown over double-line drainage shall be encoded. All opening bridges shall be encoded as return level No. 1. All other bridges shall be encoded as return level No. 3.

5.9.2.2 Hydrographic features.-

5.9.2.2.1 Lakes.- Lakes shown on the AMS series 250 charts having a minimum dimension of 2,500 feet by 1,000 feet shall be provided.

5.9.2.2.2 Waterways.- All waterways as defined in 5.9.1.2.4 shall be shown.

5.9.3 Low altitude effects.- Targets which are radar significant only at aircraft altitudes below 1,000 feet and at vertical aspect angles of less than 10 degrees, from the horizontal and which are not encoded as class 1, 2 or 3 radar return level, shall be encoded as No. 1 special effects. Examples of such targets are as follows:

(a) Isolated wooden structures or groups of structures which are a minimum of 40 feet higher than the surrounding structures or terrain

(b) Isolated vertical obstructions, such as steel television or radio towers, water towers, smokestacks, steep-sided natural formations less than the local contour interval in height and the like, which are a minimum of 200 feet in height

(c) Power transmission lines which are radar significant. Powerlines as shown on the ATC series 100/200 or, when ATC are not available, on the operational navigation charts (ONC) shall be provided. Further, only those powerlines which occur in open flat terrain or significant vegetative openings of over 300 feet will be considered radar significant and will be encoded.

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These lines shall be portrayed at return level No. 4 as 500 foot diameter circles spaced 3,000 feet between centers.

5.10 Elevation-transmittance relationships.- The elevation-transmittance relationships for the topographic image shall be as defined in 5.10.2 for type I transparencies and 5.10.3 for type II transparencies. The variation in the density of each gray level shall be held to within ± 0.035 of the specified value, and that the required density difference between adjacent density values shall be maintained within ± 0.02 density.

5.10.1 Elevation data.- Elevation contour data shall be obtained from AMS series 250 charts, from ATC series 100/200 or other sources as specified by the procuring agency.

5.10.2 Elevation data, type I transparencies.-

5.10.2.1 Elevation-transmittance slopes.- The elevation-transmittance relationships for the topographic image shall be as in table I.

TABLE I. ELEVATION TRANSMITTANCE RELATIONSHIPS

LOWER COORDINATES			UPPER COORDINATES	
Slope	Elevation in Feet	% Transmittance	Elevation in Feet	% Transmittance
A	1000	50	0	1
B	2450	50	0	1
C	4900	50	0	1
D	8000	50	0	1
E	16000	50	0	1
1(B)	16000	50	500	11
2(C)	16000	50	1000	11
3(D)	16000	50	1632	11
4(C)	24000	50	1000	11
5(C)	29000	50	1000	11

5.10.2.2 Lower coordinates.- The lower coordinates for any slope shall be the maximum elevation required, with respect to the standard slopes listed in 5.10.2.1 and 50.0 percent transmittance.

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5.10.2.3 Upper coordinates for letter slope.- The upper coordinates for the lettered slope shall be 1 percent transmittance and "0" elevation.

5.10.2.4 Upper coordinates for numbered slope.- The upper coordinates for a numbered slope shall be the intersection of the lettered slope with 11.0 percent transmittance.

5.10.2.5 Elevation-transmittance curves.- The elevation-transmittance relation for any given transparency may be any one of the slopes shown on table I.

5.10.3 Elevation data, type II transparencies.-

5.10.3.1 Elevation-transmittance relationship.- This relation shall be stored on the plate using up to 29 density values to describe all required contour levels. Maximum density shall be 2.00 (1 percent transmission) corresponding to "0" elevation. Where maximum elevation is 7280 feet or more, all 29 densities shall be used, the minimum then being 0.16 (69 percent transmission) and representing the contour level corresponding to that maximum elevation. Where maximum elevation is less than 7280 feet, the minimum number of densities shall be employed that will render the contour levels drawn. The elevation-transmittance (percent transmission) function, plotted on linear ordinates, shall be linear except that this line may have up to 4 linear connected segments each of different slope and each comprising not less than 4 contour levels.

5.10.3.2 Density value of increments.- The density value of the elevation increments shall differ, between sequentially adjacent increments as follows:

(a) Increments 1 through 19, starting with 2.00 density and ending with 0.75 density shall differ by 0.07 density

(b) Increments 19 through 26, starting with 0.74 density and ending with 0.32 density, shall differ by 0.06 density

(c) Increments 26 through 29, starting with 0.32 density and ending with 0.16 density, shall differ by 0.05 density

5.10.3.3 Elevation value assignment.- The transparency manufacturer shall submit, to the procuring agency, a report covering the assignment of elevation values as required by the governing contract.

5.11 Radar return values.- The levels of radar return shall be encoded on the transparency using the indicated values of density as shown in table II. The variation in the density of each gray level shall be held within the tolerances specified in table II.

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TABLE II. RADAR REFLECTANCE VALUES

Return	Density	% Transmittance
Water	$2.20^{+0.10}_{-0.15}$	0.63
Marsh	1.70 ± 0.10	1.90
Land	$1.13^{+0.08}_{-0.04}$	7.41
No. 1 special effects	0.74 ± 0.06	18.2
No. 3 power return	0.52 ± 0.06	30.2
No. 2 power return	0.32 ± 0.06	47.9
No. 1 power return	$0.16^{+0.04}_{-0.08}$	69.2

NOTE: Returns shown on ATC 200/100 series shall be encoded on the transparencies as follows:

(a) First intensity and fluctuating firs intensity returns-- No. 1 return

(b) Second intensity and fluctuating second intensity returns-- No. 2 return

(c) Third intensity and fluctuating third intensity, radar insignificant, radar unknown and less than 6 percent roof coverage-- No. 3 return

5.12 Transparency case.- The case shall be an appropriate instrument type which may serve as a storage case. Provisions for padding shall be included to enable the case to be used as a shipping case. The case shall have provisions to store two transparencies and appropriate documentation.

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Copies of this standard for military use may be obtained as indicated in the foreword to, or the general provisions of, the Index of Military Specifications and Standards.

The title and identifying symbol should be stipulated when requesting copies of military standards.

Preparing activity:
Navy-ONR(NTDC)

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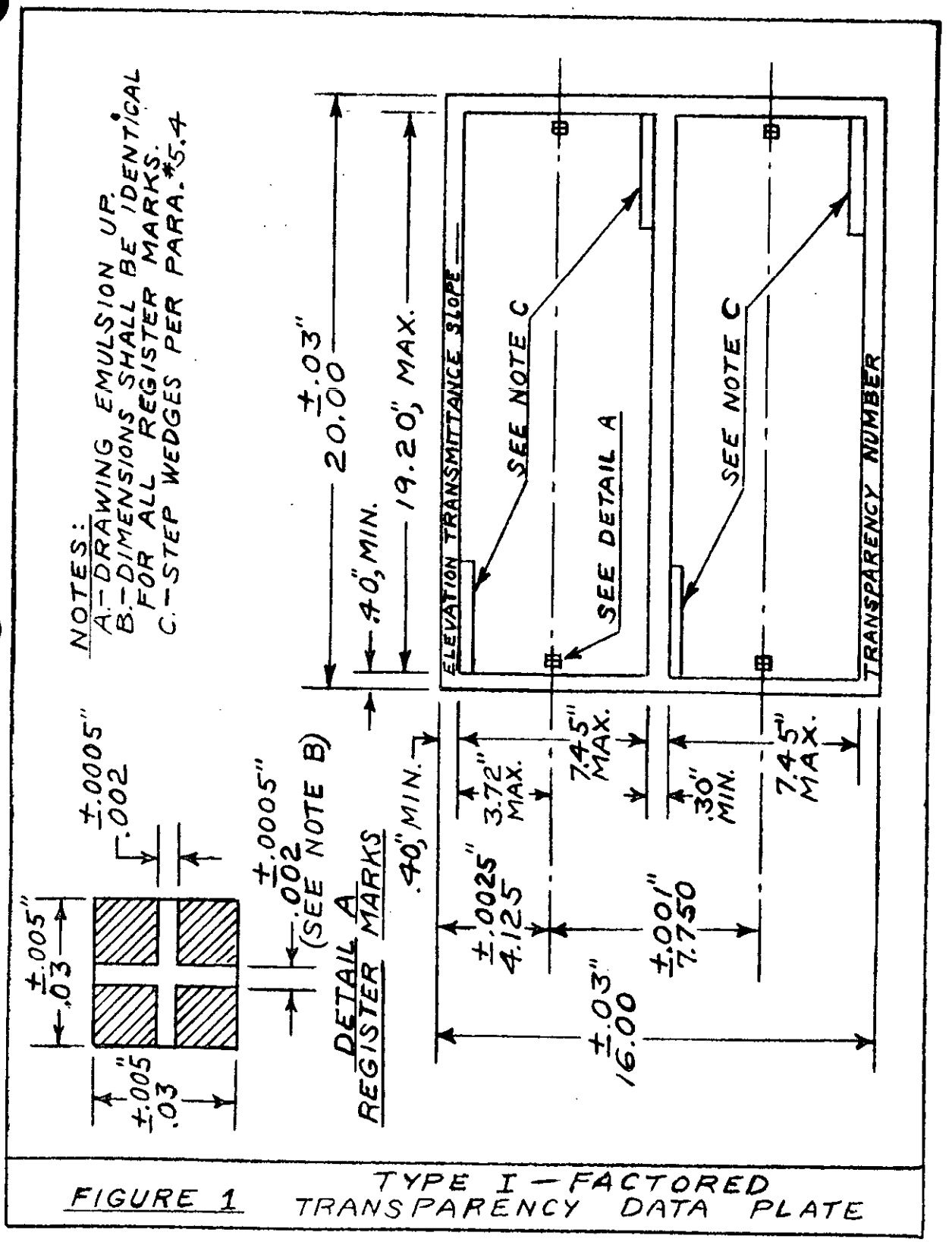


FIGURE 1 TYPE I — FACTORED TRANSPARENCY DATA PLATE

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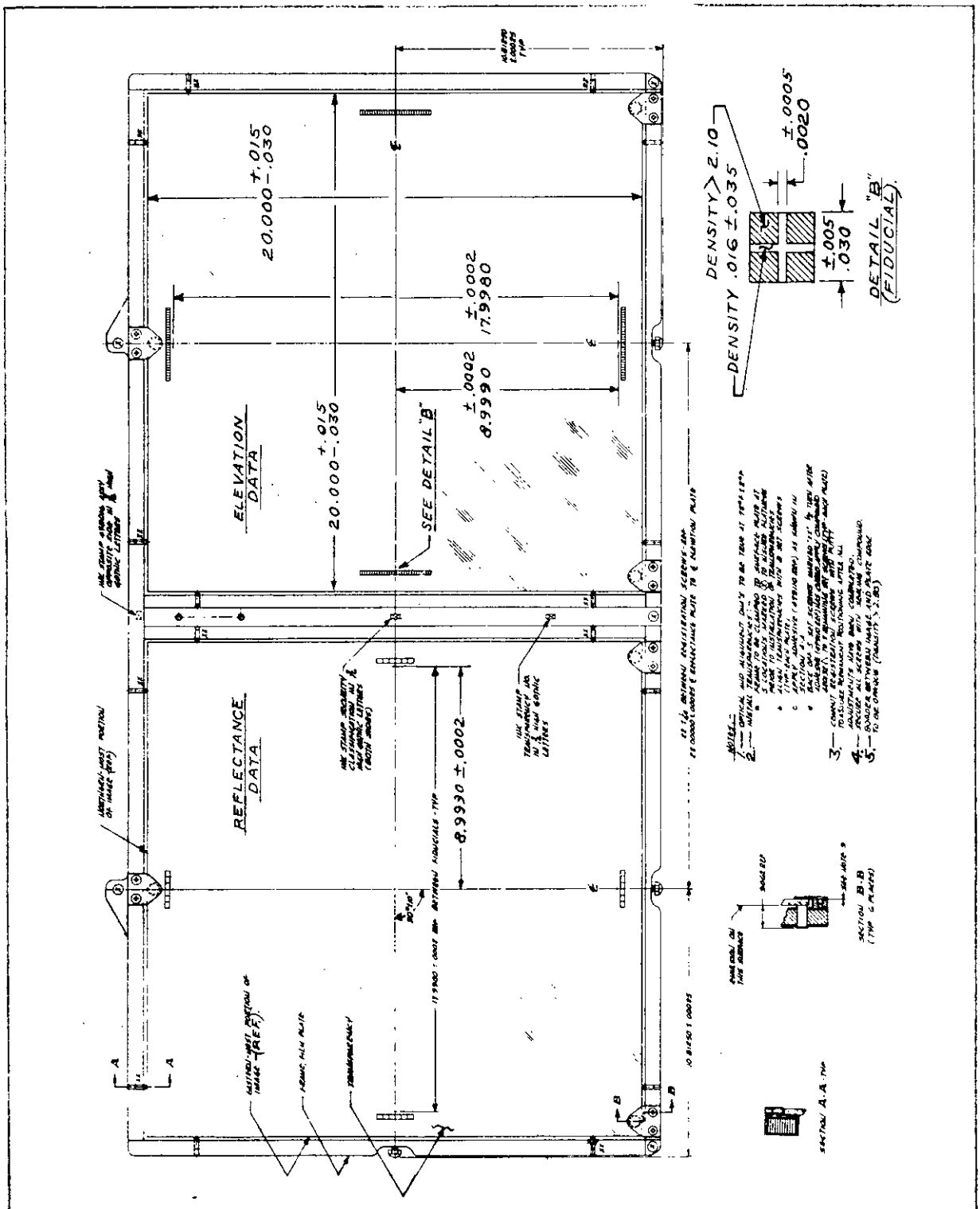


FIGURE 2 TYPE II FACTORED TRANSPARENCY DATA PLATES — ASS'Y.

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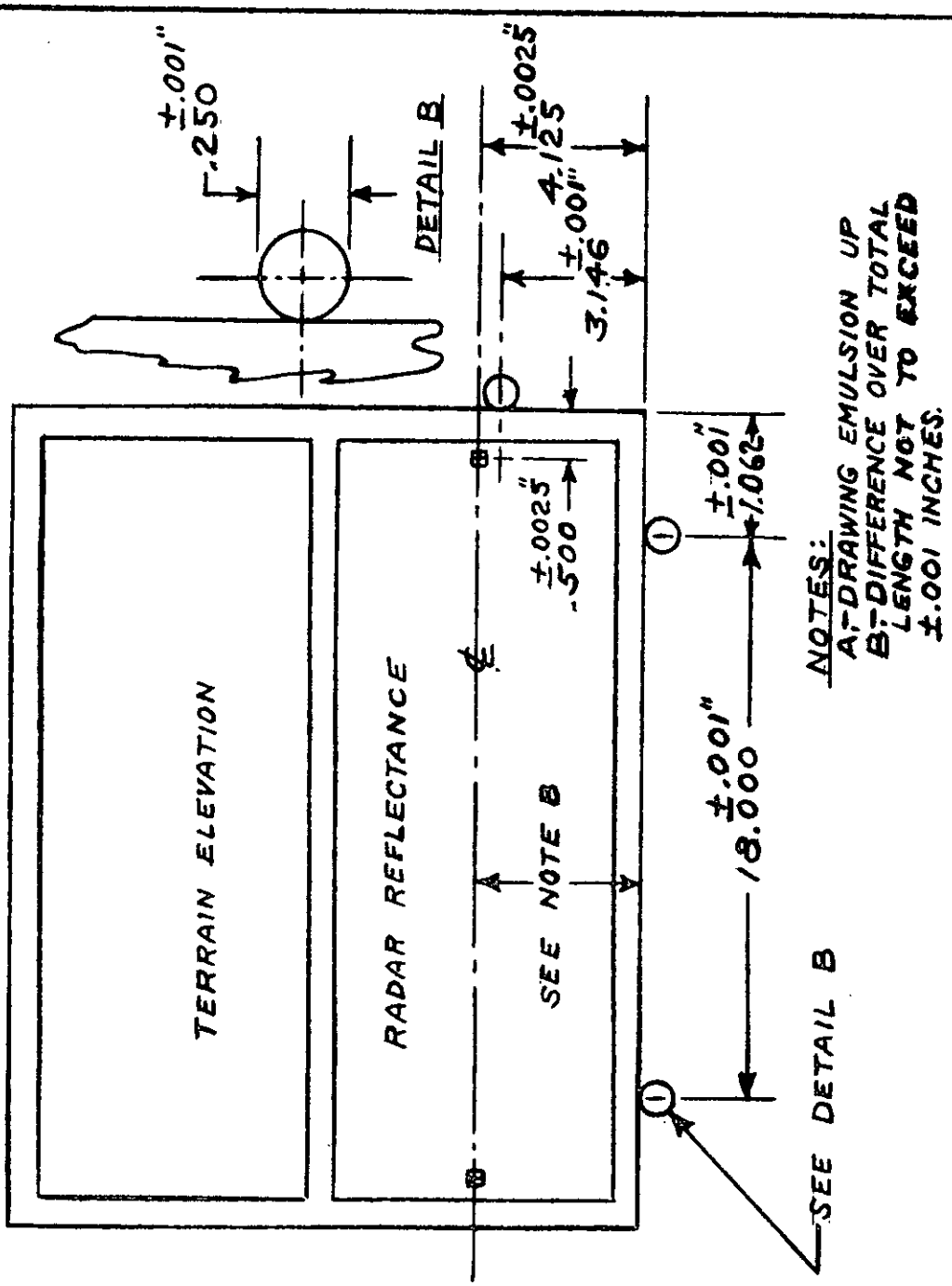


FIGURE 3 TYPE I TRANSPARENCY PLATE POSITIONING

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